



×



CLICK-A-THON 2026

· BENGALURU

· SONYLIV TRACK

Counting real viewers, not ghosts.

BACKGROUND-ONLY CONCURRENCY AT STREAMING SCALE

Count only what is **actually being watched** — exact peaks, corrections that are arithmetic instead of rebuilds, and verification that fails loud. Every number in this deck is from the **unseen day**, measured on the live service.

● **fastandfurious.live** — the concurrency curve, live

● **chat.fastandfurious.live** — ask it in plain English

ENGINE	UNSEEN DAY	PEAK
ClickHouse Cloud 26.2 · ap-south-1 · 2 replicas	7,000,000 events · 108,486 sessions	19,882 concurrent · 2026-07-31 11:15 UTC

A session boundary is not a viewer.

Counting every session between its first and last event is the obvious implementation, and on the unseen day it is wrong by a fifth. Phones get pocketed, players get paused, streams stall — all of it keeps a session technically open and none of it is an audience.

NAIVE · SESSION BOUNDARIES

24,069

peak concurrent sessions, 2026-07-31 11:16 UTC.
1,453,136 aggregate session-minutes.

MEASURED · FOREGROUND-ONLY

19,882

peak concurrent sessions, 11:15 UTC — a different minute.
1,117,802 aggregate session-minutes.

OVERCOUNT AT PEAK

+21.1%

Peak-vs-peak. The two peaks do not land on the same minute, so no correction factor fixes it after the fact.

OVERCOUNT IN AGGREGATE

+30.0%

335,334 session-minutes of audience that was never watching anything.

AT THE SCALE IT MATTERS

4,187

phantom viewers at the single busiest minute of the day — the number ad load and capacity are sized against.

We profiled first.

Then the unseen day moved the ground.

Every load-bearing claim, measured on both datasets. The tuning day taught us some lessons that were **true only of the tuning day** — which is exactly why nothing in the pipeline is allowed to assume them.

THE DICTIONARY SAYS	WHAT IS ACTUALLY TRUE	TUNING DAY	UNSEEN DAY
Heartbeat every 60 s	The periodic ping is a trio at identical millisecond timestamps, and it fires every 40 s on both days.	p50 40.0 s	p50 40.0 s unchanged — the one stable fact
A heartbeat proves playback	iOS does not emit the trio. Trio-based liveness silently drops most Apple traffic.	74.4% IPHONE, no trio	58.6% / 62.5% / 63.6% IPHONE · IPAD · APPLE_TV
Pause is an event type	event_type= 'VideoHeartbeat ' is a grab-bag. Pause exists only as lowercase ' pause ' inside it.	41 names	42 names a new one, golive, appeared
Every session closes	The tuning day lied. There, all 10,866 sessions had a Start and an End. On the unseen day a third never close, and a quarter were already running when the window opened.	0% open sessions	34.7% open 37,649 never end · 25,403 never start
bg/fg are “not guaranteed”	The dictionary was right and the tuning day was not. There, 100% of sessions carried both. Here, most carry neither — so backgrounding cannot be the only exclusion rule.	100% carry both	6.51% carry both · 86.4% no bg at all

The brief poses five questions.

Here they are, answered.

“THE QUESTIONS THIS PROBLEM TURNS ON”		OUR ANSWER	MEASURED, UNSEEN DAY
1	How do you define an active interval when the heartbeat is missing, the player is paused, or the app is backgrounded?	A five-clause predicate in a versioned <code>policy.yaml</code> : started $\wedge$ ¬terminated $\wedge$ foreground $\wedge$ playing $\wedge$ live within 120 s . Background is a hard edge, pause stops playback, error stops it without ending the session.	0.2% of legitimate gaps falsely split at $T = 120\text{ s}$
2	How should active ranges be represented — interval arrays, normalized intervals, pre-aggregated minute deltas, or a hybrid?	Hybrid, and both views the brief asks for . Normalized half-open intervals are the session-aware source of truth; signed ± 1 deltas at their edges are the serving form; a dense minute tier sits on top. A session-independent <code>uniqCombined64</code> estimate runs alongside as the comparison path.	7,000,000 $\rightarrow$ 163,740 events to intervals – 43× fewer rows
3	How do you compute accurate minute-wise peak and average without scanning raw session history on every query?	Prefix-sum the deltas from a day anchor, never from $t = 0$. Average is time-weighted over exact constant-concurrency segments, not sampled at minute boundaries. Hour and day peak is <code>max(minute_peak)</code> ; average is <code>sum(active_ms) / duration</code> .	5 ms / 8,192 rows vs 476 ms and 13.9 M rows recomputing it raw
4	How does the model stay filter-friendly across platform, country, content, video type and time grain?	A <code>rollup_mask</code> bitmask materialises nine dimension combinations inside the sort key. Peak is always computed after aggregating the requested combination — never by summing per-dimension peaks, which land on different minutes.	1,100 @ 11:23 SAMSUNG_HTML_TV's own peak; only 1,054 at the global one
5	How do you handle sessions that are still open , whose active ranges keep growing as new heartbeats arrive?	Two mechanisms. “Right now” is a separate $O(\text{open sessions})$ table where a quiet session simply falls out of the WHERE clause — nothing is written when a viewer stops. For history, a changed session publishes new – old boundary maps as signed corrections.	37,649 sessions that never close – 34.7% of the day

Where the statement is ambiguous, we answer in a **file**, not a comment.

Every semantic switch lives in a versioned `policy.yaml`, is stamped onto every published answer, and has a measured cost for flipping it.

```
-- policy: sonyliv-active-v1
active = started
  AND NOT terminated
  AND foreground
  AND playing
  AND event_time < last_signal + 120s
```

WHY 120 SECONDS

Three missed 40-second ticks, and the cadence held at **p50 40.0 s** on the unseen day too. A 120 s lease falsely splits about **0.2%** of continuous playback; 180 s recovers a few hundredths of a point more while adding a full minute of phantom viewing after every genuine stop. The knob is configurable and both alternatives are measured.

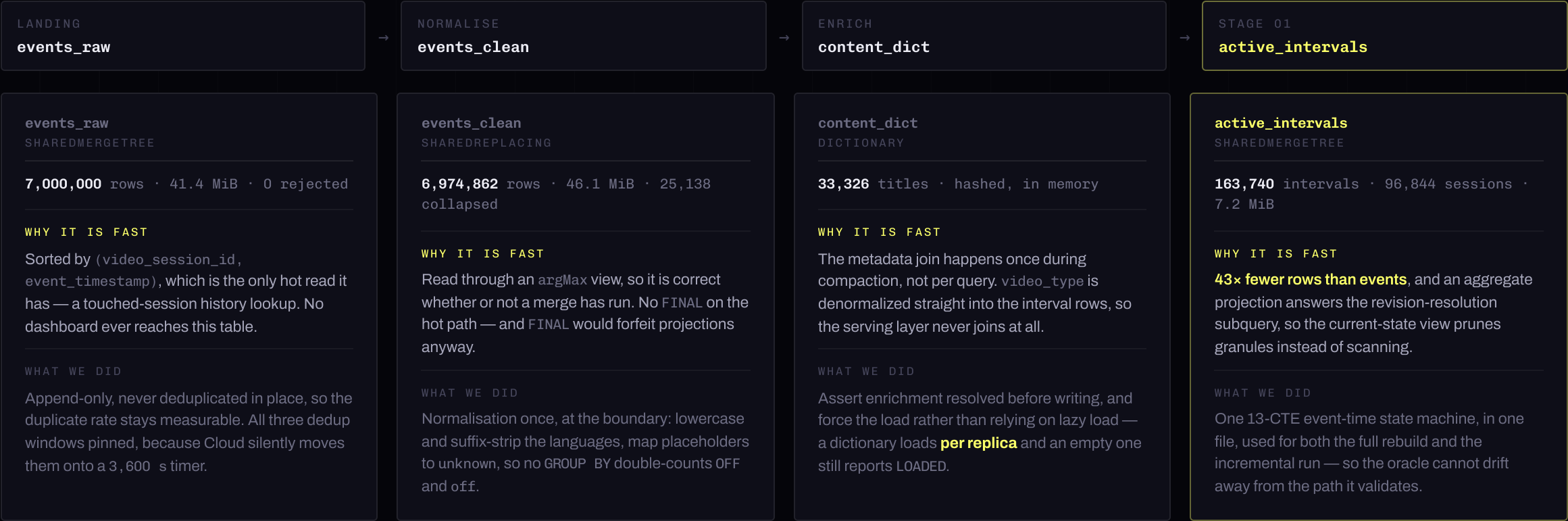
EVERY CLAUSE IS DEFENDED BY A MEASUREMENT

Liveness = <i>any</i> event	not the periodic trio — that drops 58.6% of iPhone and 63.6% of Apple TV sessions
Background is a hard edge	heartbeats stop within ~5 s of the bg edge; waiting out the lease adds up to 120 s of nothing
Paused counts as inactive	“count only truly active <i>playback</i> intervals”. A paused stream serves no ads. 98,703 pause events on the unseen day — it is not a rounding error
The lease carries the day	with only 6.51% of sessions emitting bg/fg at all, silence is the primary exclusion signal, not the backup one
UTC end to end	an IST rendering moves events across a calendar boundary and would change daily partitions. Display-time conversion only

WHY THIS SURVIVED THE UNSEEN DAY

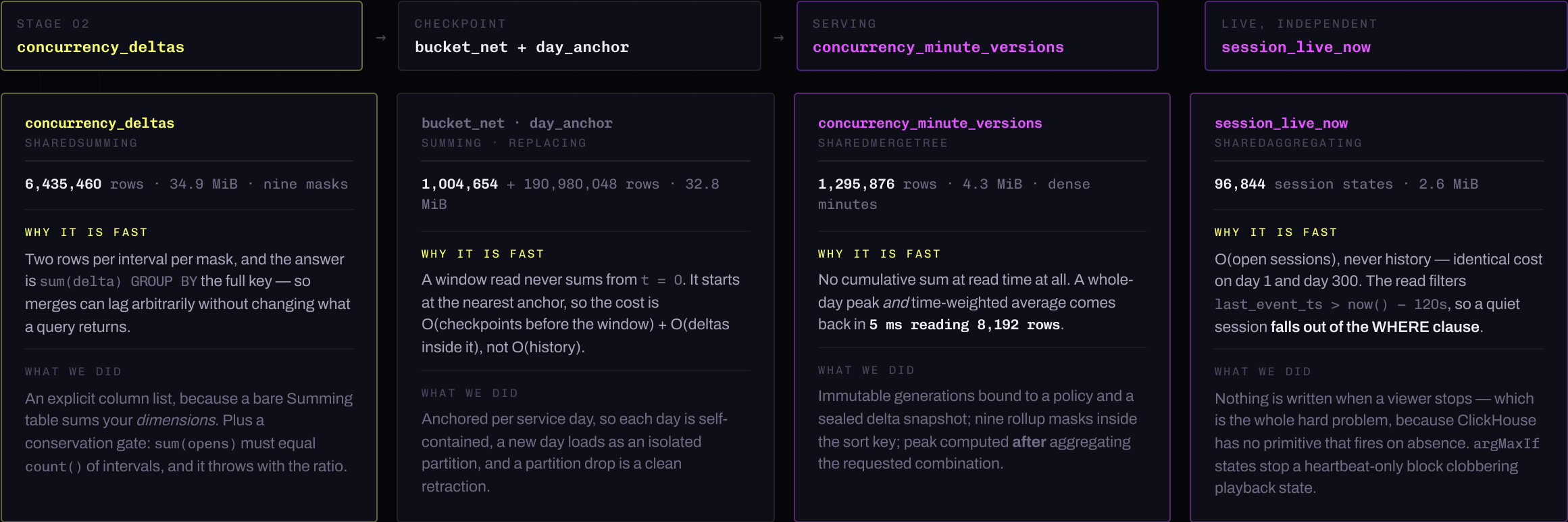
The predicate never assumed a session ends, never assumed a background event arrives, and never assumed the trio exists. Each of those three assumptions would have been safe on the tuning data and **wrong here**.

Seven million events in. 163,740 intervals out.

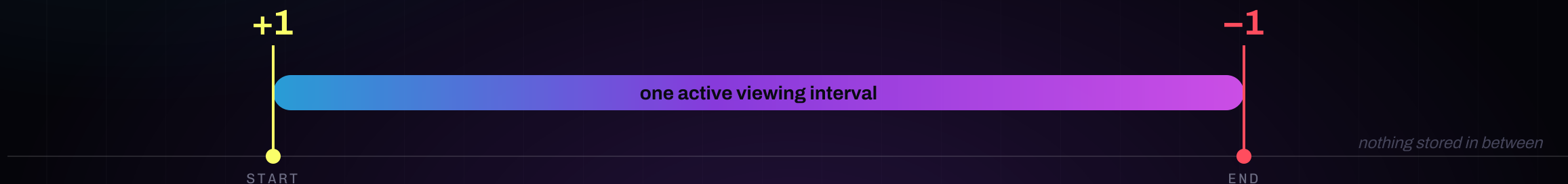


Two questions. Two structures.

Conflating them is the whole mistake.



Two rows. That is the whole idea.



Concurrency at any instant is the running sum of those edges. Storage is bounded by **sessions**, never by session-minutes — and a correction becomes **subtraction** instead of a rebuild.

EVENTS IN

7,000,000

→

ROWS WE KEEP

163,740

→

ROWS A DASHBOARD READS

8,192

→

TIME TO ANSWER

5 ms

$\text{CORRECTION}(T) = \text{NEW_BOUNDARY_MAP}(T) - \text{OLD_BOUNDARY_MAP}(T)$ · A LATE PAUSE RETRACTS ITS OWN TAIL – NO ALTER UPDATE, NO RE-READ OF HISTORY

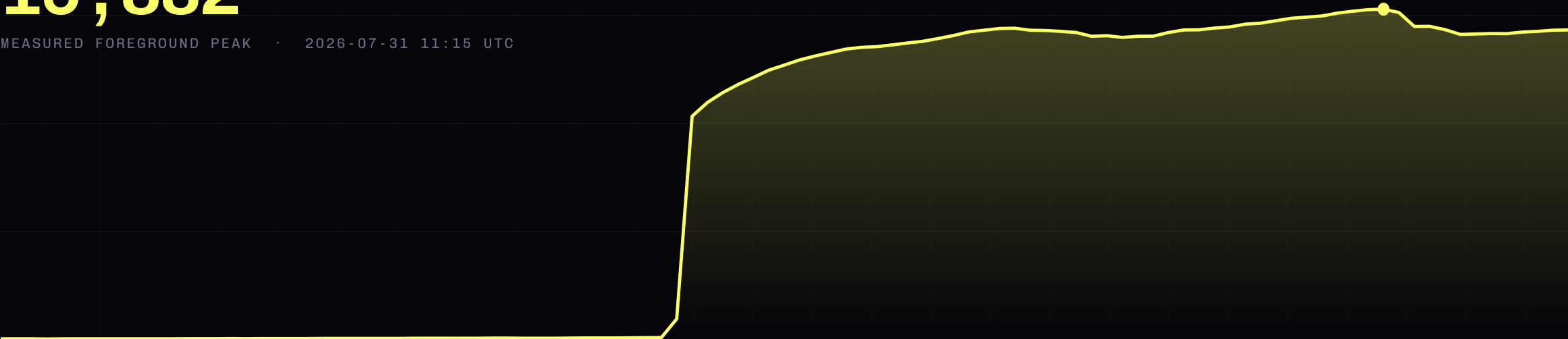
SONYLIV × CLICKHOUSE

The unseen day, answered.

24,069 naive session-boundary peak — what session boundaries claim

19,882

MEASURED FOREGROUND PEAK · 2026-07-31 11:15 UTC



09:45 UTC · 2026-07-31 · one point per minute, straight off the serving tier · 11:27 UTC

PEAK · MINUTE GRAIN 19,882 2026-07-31 11:15 UTC	PEAK · HOUR GRAIN 19,882 / avg 9,911.6 11:00 UTC — average is time-weighted, not sampled	CONSERVATION GATE 1.000 sum(opens) = sum(closes) = count() = 163,740	SLICE PEAKS MOVE 1,100 @ 11:23 SAMSUNG_HTML_TV — only 1,054 at the global peak minute
--	---	---	--

A third of the day never closed. It did not matter.

On the tuning data every one of 10,866 sessions had a Start and an End, so the open-session problem could only be simulated. The unseen day supplied it for real — **37,649 sessions that never end** and **25,403 that were already running** when the window opened.

SESSIONS THAT NEVER CLOSE	SESSIONS WITH NO START	INTERVALS PUBLISHED	REBUILDS REQUIRED
37,649	25,403	163,740	0
34.7% of the day — the tuning extract had zero	the window opens mid-stream; state is inferred, not assumed	over 96,844 sessions, all revision-resolved	every change is an additive, signed row

WHY AN OPEN SESSION IS NOT A SPECIAL CASE

An interval whose session is still live is published with a **provisional** end at the lease expiry. When the next heartbeat lands, the session is re-derived and the pipeline publishes the *difference*: **+1** at the old expiry, **-1** at the new one.

The curve is therefore always correct as of its watermark, and it converges without anything being rewritten. The touched-session query is bounded — it reads that session's history, not the day's.

PROVED ON A HOSTILE FIXTURE, BEFORE THE DAY ARRIVED

Event-time replay, 1% held back 30 min	0 diffs
One late pause → signed rows emitted	320
Signed sum of those rows	0
Independent paths that agree on active-ms	3
Reusing a batch ID hard-fails. Reusing a minute generation hard-fails. A partial-minute request is rejected and routed to the exact endpoint query rather than quietly answered with a truncated bucket.	

The number that matters is not latency. It is **rows read**.

The brief says judges will look at what the queries read, not just how fast they return. Both columns below come from the same service, on the same 7-million-event day.

BENCHMARK SHAPES · UNSEEN DAY		
QUERY	ELAPSED	ROWS READ
Day peak + time-weighted average	5 ms	8,192
Per-platform peak and each platform's own peak minute	7 ms	8,192
Hour-grain peak + average, all hours	5 ms	8,192
Conservation gate over every delta row	110 ms	677,016
Naive concurrency, recomputed from events the thing the serving tier exists to avoid	476 ms	13,945,916
95× faster and 1,700× fewer rows than recomputing the same answer from events — and that gap widens with every day of history, because one side is bounded by dimensions and the other by events.		

ONE REWRITE, MEASURED ON THE LIVE SERVICE		
SELECT max(event_ts) FROM events_dedup was the #1 SELECT by both total time and total bytes — 694 executions reading 3.13 billion rows and 56.3 GB. event_ts is itself a GROUP BY key of that view, so grouping cannot change its maximum. The base table is provably identical.		
	ELAPSED	BYTES READ
FROM events_dedup	1.058 s	162,146,488
FROM events_clean	0.104 s	72,000,384
	10.2× faster	2.25× fewer

INGESTED

7,000,000

events, 0 rejected

ON DISK

87.45 MiB

raw + clean, 35.2× compression

SERVING P50

7 ms

from system.query_log

Deployed, not described.

The database comes up from nothing with one command, and the command refuses to pass quietly. That is how the unseen day was loaded.

```
$ ./scripts/bootstrap.sh
```

Creates every table, view, materialized view and dictionary; loads the seed CSVs; builds the interval and serving tiers; then runs **seven assertions** that are all reference-free — none contains a number from the tuning day. Python stdlib over HTTPS; `clickhouse-client` is not required.

SERVICE

sonyliv
aws ap-south-1
ClickHouse 26.2
2 replicas

UNSEEN-DAY RUN

7,000,000 loaded
0 rejected
163,740 intervals
ratio 1.000

WHAT WE WOULD NOT CLAIM

Compactor coordination	A fenced Keeper lease is specified but not deployed. The embedded verifier is single-worker and does not prove that external property.
The mask set	A missing mask's <i>deltas</i> can be re-derived from mask 15; its <i>peak</i> cannot, because peaks do not sum. The benchmark query set was never published.
Latency on other hardware	Never claimed without a <code>system.query_log</code> row from the target service behind it.

BEHAVIOUR AT 100×

700 M events a day, ~10.8 M sessions, ~16.4 M intervals. The serving tier grows with **dimensions × minutes**, not with events: 1,440 minutes × nine masks is the same shape at 1× and at 100×. What grows is the interval build, and that is a partitioned, per-day, restartable job — not a query anyone waits on.

Nine ways to be confidently wrong

— none of which reproduce on a laptop.

01	A dictionary reports <code>LOADED</code> with zero rows , per replica. Every <code>dictGetOrDefault</code> silently returns the default; routing decides whether you get the right answer.	33,464 / 0
02	<code>MergeTree</code> becomes <code>SharedMergeTree</code> on Cloud, moving dedup onto a window that expires on a timer . “Re-running a load is a no-op” stops being true one hour in.	3,600 s
03	<code>insert_deduplication_token</code> did not make a large <code>INSERT SELECT</code> idempotent. It ran twice, four seconds apart, byte-identical.	524,292 ×2
04	A doubled Summing curve passes every balance invariant. <code>sum(net)=0</code> , <code>min(running)=0</code> , <code>opens=closes</code> — all still true at exactly twice the peak.	exactly 2×
05	<code>SummingMergeTree</code> keeps the oldest <code>DateTime64</code> , not an arbitrary one. A lease timestamp therefore reads as already expired.	91.05%
06	With no explicit column list, <code>Summing</code> sums your dimensions . A wrapped <code>user_key</code> still looks plausible; a corrupted <code>content_id</code> misses the dictionary entirely.	96.7% > 2 ⁶⁴
07	An output alias shadows the column it filters on . The predicate becomes <code>13 = 5</code> and the deployed view returned zero rows, with no error.	0 vs 85,553
08	Adjacent SQL string literals are a parse error, so every gate in the minute tier's verifier had never executed — rows shipped with no verification firing.	6 gates dead
09	Projections throw on Replacing / Summing / Aggregating tables by default, and a <code>SELECT ... FINAL</code> cannot use one at all.	7 of 13 blocked

AND THE RULES THEY TURNED INTO

A check that only inspects one layer cannot validate that layer.

Internal-consistency checks are structurally incapable of detecting a scalar multiple. Every stage carries a conservation check tying its output back to its *input* — on the unseen day, `sum(opens)` in the deltas equals `count()` of intervals at ratio **1.000**.

No gate may contain a number we happen to know.

Gates that hardcoded tuning-day answers were found and removed. Had they survived, every one would have fired on *correct* unseen-day output and blocked the pipeline — 19,882 is not 2,305.

Assert enrichment resolved before writing.

A dimension that is 100% fallback is a failure, not a data characteristic. This holds whatever the root cause turns out to be.

Run the verifier. Then break something on purpose.

A verification file that has never been executed is worse than no verification, because its presence is read as coverage.

Measure on the service. Never infer.

Every defect above was invisible on a laptop and invisible in code review.

Two integrations, each doing work the pipeline actually needs.

CLICKSTACK

It observes the pipeline itself.

One trace spans the whole publication path:

```
replay.read → raw.insert → session.normalize → boundary.correct  
→ minute.build → generation.validate → generation.publish
```

Carrying `run_id`, source SHA, policy version, touched sessions, ingest-lag quantiles, correction backlog, dictionary misses, read rows/bytes and the answer hash — the **pipeline evidence bundle** the unseen day demands.

ALERTS THAT ARE ACTIONABLE

Any negative concurrency · non-zero daily delta balance · dictionary misses above zero · generation hash mismatch against the independent oracle.

Concurrency decline — the brief's own suggested LLM + ClickStack case — comes off the same minute tier. A drop is trivial to detect and useless without a cause, so the alert carries the slice that moved and the model separates *the asset ended* from *the platform broke* from *nobody is enjoying this*.

LIBRECHAT + CLICKHOUSE MCP

It changed the schema, not just the demo.

"What was peak concurrency on Android in the last hour?" answered conversationally, straight off the serving tables.

The honest reason `concurrency_minute_versions` exists is this integration: a flat, dense, one-table read surface is what makes text-to-SQL answerable. An LLM can reliably write:

```
max(minute_peak) ... WHERE minute_start  
BETWEEN ... AND ...
```

It will **not** reliably write a day-anchored cumulative sum over millisecond boundaries. Latency was a side effect, never the justification.

Nine rollup masks materialised — 0, 1, 2, 3, 4, 5, 8, 9, 15. Mask 12 is omitted because `video_type` is functionally determined by `content_id`, and mask 12 matched mask 4 on every row tested.

● **fastandfurious.live** — the curve builds live; filter it and the minute view answers instantly

● **chat.fastandfurious.live** — then ask the follow-up in plain English

A wrong number costs the same as a crash,
and it arrives **without warning**.
So we built the thing that **fails loud** instead.

● [fastandfurious.live](#)

● [chat.fastandfurious.live](#)

UNSEEN-DAY PEAK	CONSERVATION	ANSWER TIME
19,882 @ 11:15	ratio 1.000	5 ms · 8,192 rows

JUDGED ON FIVE AXES — WHERE EACH ONE LIVES

Correctness	foreground-only excludes pause, background and silence; conservation gate ties deltas back to intervals at ratio 1.000
Query performance	8,192 rows for a whole-day peak and average — 1,700× fewer than recomputing it, with a <code>system.query_log</code> row behind every figure
Update handling	34.7% of the unseen day never closed; signed corrections absorbed it with zero rebuilds
Design quality	every switch versioned in <code>policy.yaml</code> with its measured cost; every rejected alternative carries the number that rejected it
The unseen day	full-scan and incremental are the same file, so the oracle cannot drift from the path it validates

`ingest/` Go write path · `pipeline/` the deployed stages · `solution/` policy, SQL and the hash-locked harness · `optimizations/` measured rewrites · `scripts/` one-command bring-up



×



CLICK-A-THON 2026 · SONYLIV TRACK

MIT LICENSED · PUBLIC REPOSITORY · COUNTING REAL VIEWERS, NOT GHOSTS